

Chapter 1: Foundations of Machine Learning

Course material

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① 1.1 History and evolution of machine learning

- From least squares (1805) to Generative AI
- The “AI winters” and why they happened

② 1.2 Fundamental concepts

- Data, features, and labels
- *Overfitting*, cross-validation, and metrics
- The ML *pipeline* end to end

These foundations underpin everything that follows: without them, it's hard to understand why a model works or fails.

Why learn machine learning?

- In **traditional** programming, a human writes every rule by hand:
 - `if temperature > 30: turn_on_ac()`
- But many problems are too complex to describe with fixed rules
- **Machine learning inverts the paradigm**: instead of writing the rules, we teach the machine to *learn* them from examples

Example: no one can hand-write a rule that tells a cat from a dog in a photo, but a model can *learn* one from thousands of already-labeled photos.

The new programming paradigm

- For a machine to learn, **three elements** are needed:
 - ① Input data
 - ② Examples of the expected output (**labels**)
 - ③ A way to measure whether it's doing well (**loss function**)
- AI isn't a new idea: it coexists with symbolic AI (hand-written logical rules), which dominated the field from the 1950s to the late 1980s

Mathematical roots and the Turing Test (1805-1950)

- 1805: Legendre and Gauss publish **least squares** — the earliest form of linear regression
- 1936: Fisher proposes linear discriminant analysis
- 1943: McCulloch and Pitts present the first model of an artificial neuron
- 1950: Alan Turing publishes "*Computing Machinery and Intelligence*" and introduces the **Turing Test**



Source: Elliott & Fry (1951), public domain

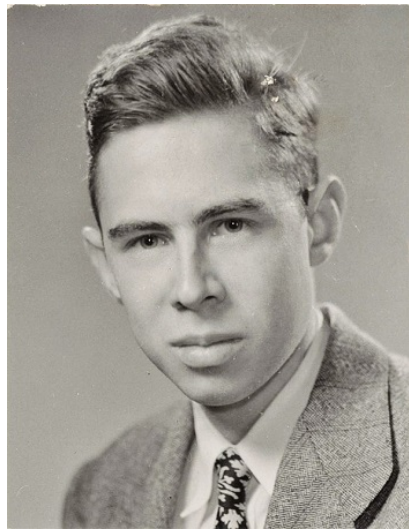
The formal birth and the symbolic era (1956-1980)

- 1956: John McCarthy organizes the **Dartmouth workshop** — AI is formally born as a research field
- For decades it was believed that hand-coding a massive set of logical rules would be enough (**symbolic AI**)
- This approach peaked with **expert systems** in the 1980s

Example: a 1980s medical expert system contained thousands of hand-written rules like “if fever and cough and chest pain, then suspect pneumonia.”

The Perceptron and the AI winters

- 1958: Frank Rosenblatt creates the **Perceptron**, the first trainable artificial-neuron model
- **First winter** (1970s): the Perceptron cannot solve non-linear problems; enthusiasm collapses
- **Second winter** (1990s): expert systems prove expensive to maintain, hard to scale, and limited
- ML truly takes off in the 1990s, driven by faster hardware and more available data



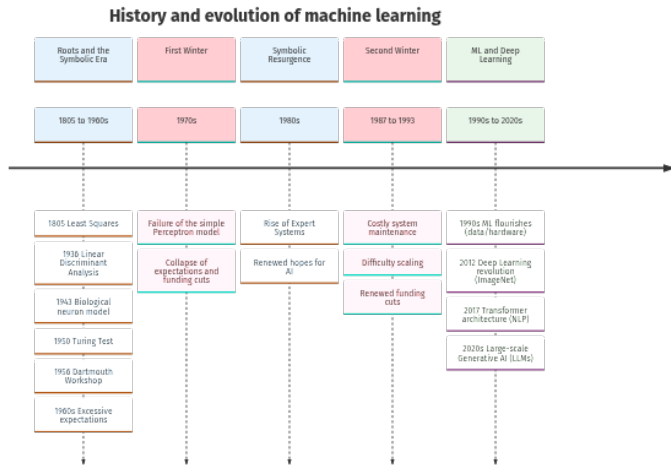
Source: Heinz Nixdorf MuseumsForum,

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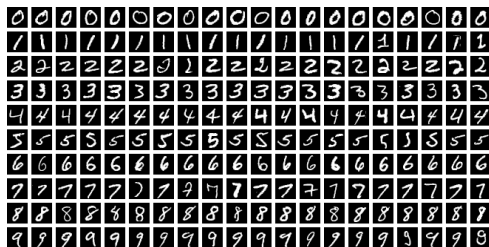
Full timeline

Cycles of “summer” (breakthroughs, funding) and “winter” (disillusionment, funding cuts).



From *Deep Learning* to the LLM era (2010-today)

- **Deep Learning** learns representations in successive layers, each more useful for the task
- 2012: Hinton's victory in the **ImageNet** challenge marks the turning point
- **MNIST** (handwritten digits 0-9) becomes the classic benchmark
- 2017: the **Transformer** architecture revolutionizes natural language processing
- 2020s: **Generative AI** (ChatGPT, Gemini) scales to billions of parameters



Source: Suvanjanprasai (2024), Wikimedia Commons, CC BY-SA 4.0

Summary: 1.1 History and evolution

- ML was born from one question: can machines learn?
- It has gone through cycles of hype and disillusionment
- Recent progress comes from more data, better hardware, and new algorithms
- Deep Learning and LLMs are the current state of the art

Data, features, and labels

Sample	Feature 1	Feature 2	Feature 3	Label
1	25	70	175	Cat
2	8	4	90	Cat
3	35	80	180	Dog

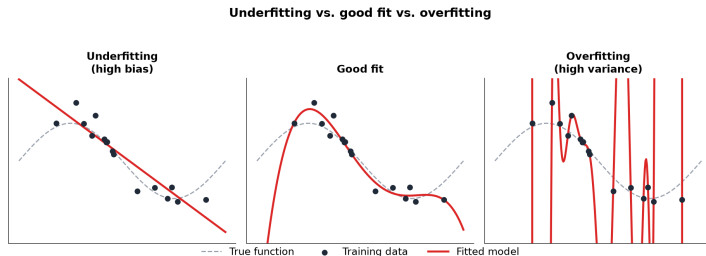
- **Sample:** a single data point (one row of the table)
- **Features:** the attributes that describe the sample
- **Label:** what we want to predict — the “ground truth”

Example: to predict a house's price, the features would be m^2 , number of rooms, and location; the label would be the price.

Overfitting vs. underfitting

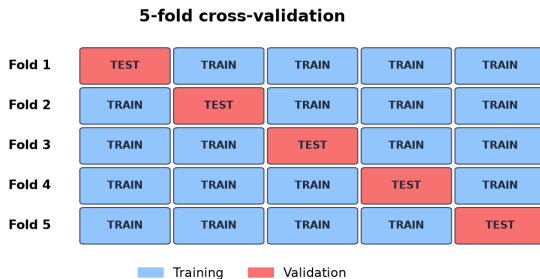
- **Underfitting** (high bias): the model is too simple, misses the real pattern
- **Overfitting** (high variance): the model memorizes training noise instead of learning the trend
- The goal is the bias-variance trade-off that generalizes best

Example: it's like studying for an exam by memorizing the exact answers of a mock test (overfitting) instead of understanding the topic: you'll fail as soon as the questions change.



K-fold cross-validation

- Standard split: **training** (fit weights), **validation** (choose hyperparameters), and **test** (final, unbiased evaluation)
- K -fold: split into K parts; train K times, each time using a different part as validation
- The final metric is the average of the K results — reduces the variance of the estimate
- **Stratification**: each partition must preserve the class proportions of the original set



Confusion matrix and metrics

Confusion matrix

		Prediction	
		Positive	Negative
Actual class	Positive	TP True Positive	FN False Negative
	Negative	FP False Positive	TN True Negative

$$\text{Precision} = \frac{TP}{TP + FP}$$

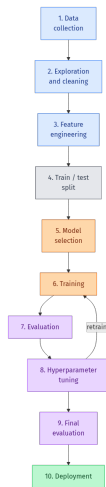
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = 2 \cdot \frac{P \times R}{P + R}$$

Example: in a medical diagnosis, a **false negative** (telling a sick patient “healthy”) is usually more serious than a false positive. That’s why recall is sometimes prioritized over precision.

The ML *pipeline*: 10 stages

- An **iterative** process: evaluation can send you back to earlier stages
- Much of the real effort lies in stages 2-3 (data) and 7-8 (tuning)
- Deployment isn't the end: it requires continuous monitoring, since real-world data changes over time (*concept drift*)



Summary: 1.2 Fundamental concepts

- **Sample / feature / label**: the basic building blocks of any dataset
- **Overfitting / underfitting**: the central tension of machine learning
- **Cross-validation**: evaluating without bias when data is scarce
- **Metrics**: pick the right one for what matters (precision or recall)
- **ML *pipeline***: 10 stages, from data collection to deployment

End of Chapter 1

Next: Chapter 2 — Supervised Learning